



NITROGEN

Q. Give a self-explanatory diagram for the prepⁿ of NH_3 gas in the lab. Mention the names of four drying agents and give reason why only lime tower is used for drying ammonia.
How NH_3 reacts with: (1) H_2 (2) Na (3) O_2 (4) CuO
Give conditions of reactions.

Action of heat on: 1. Sodium nitrate (or Potassium nitrate)

2. Ammonium nitrate.

3. Mixture of Ammonium chloride and Sodium nitrate.

4. Sodium nitrite and Ammonium nitrate

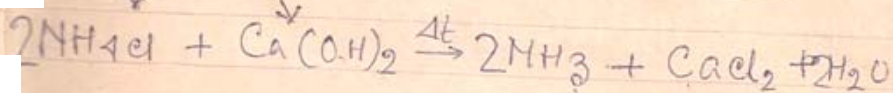
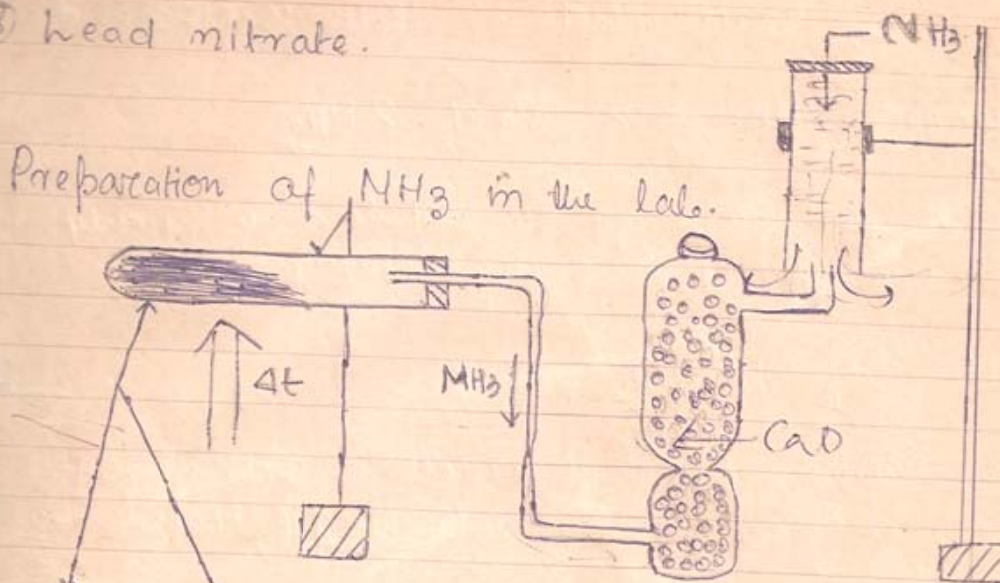
5. Mixture of Sodium nitrate and Ammonium Sulphate or Chromate

Mixture of Ammonium chloride and $\text{K}_2\text{Cr}_2\text{O}_7$



⑧ head nitrate.

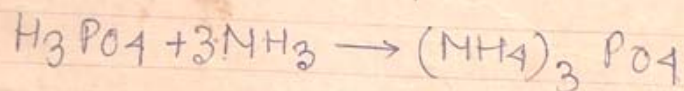
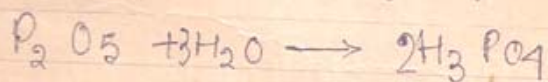
Preparation of NH_3 in the lab.



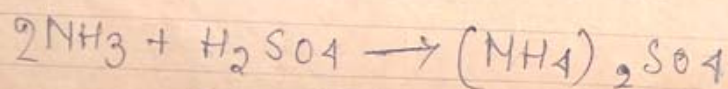
Drying agents :

- (1) P_2O_5 solid.
- (2) Conc. H_2SO_4 .
- (3) fused CaCl_2 .
- (4) CaO .

P_2O_5 is not used for drying ammonia as water present in NH_3 form phosphoric acid which again reacts with NH_3 giving ammonium phosphate $(\text{NH}_4)_3\text{PO}_4$.

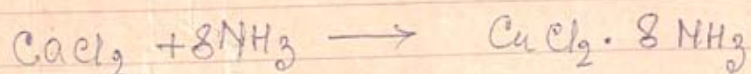


conc. H_2SO_4 is not used as NH_3 reacts





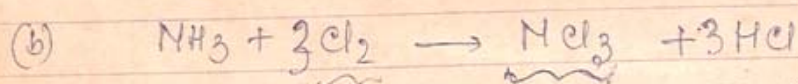
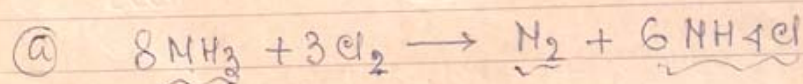
fused CaCl_2 is not used as NH_3 is absorbed.



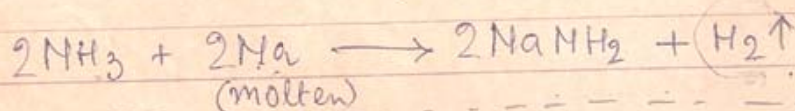
~~These is the res~~ There are the reactions why only lime tower used for drying NH_3 gas.

Reaction of NH_3 with

- ① $\text{Cl}_2 \rightarrow$ (a) When ammonia is excess we get nitrogen
(b) When Cl_2 is excess we get HCl_3 (explosive)

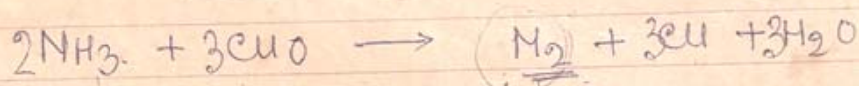


② With Na $\rightarrow$ When NH_3 gas is passed through molten Na we get sodamide and Hydrogen.



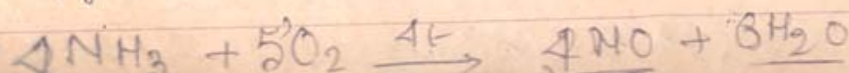
N.B This is the proof that NH_3 contains H_2 .

③ With CuO — When NH_3 gas is passed over heated CuO we get N_2 .



N.B This is the proof that — ammonia contains

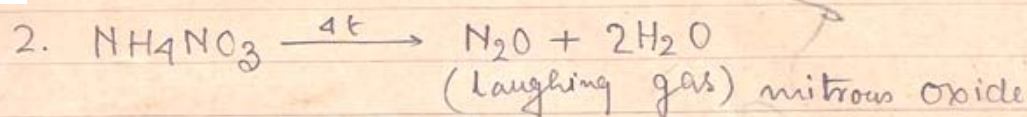
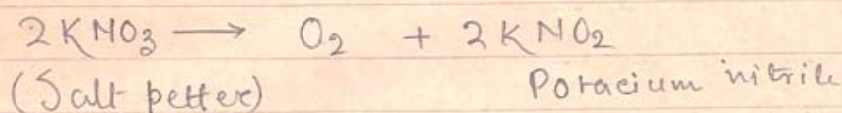
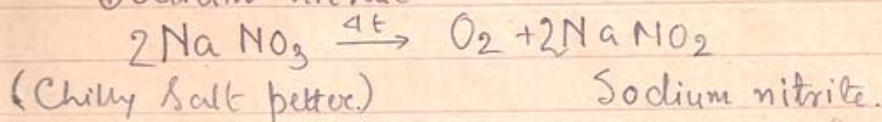
(A) With $\text{O}_2 \rightarrow$ When NH_3 is heated in the presence of Oxygen and Pt catalyst we get HNO



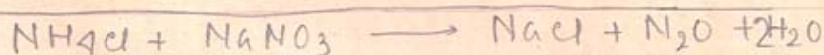
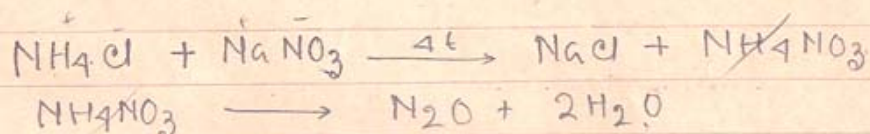


mp Action of Heat.

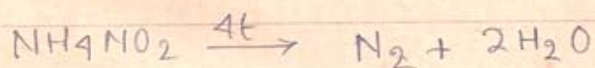
1. Sodium nitrate



3. Mixture of Ammonium chloride & Sodium Nitrate

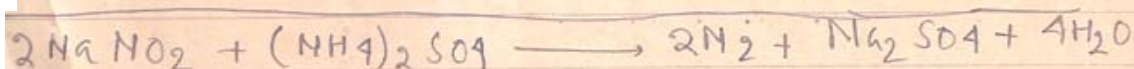
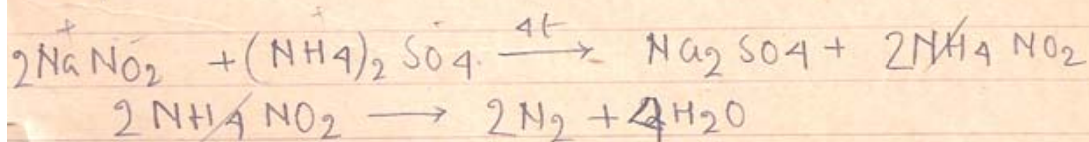


4. NH_4NO_2 (Ammonium nitrite)



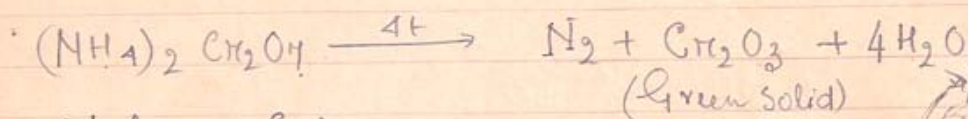
lab preⁿ of pure nitrogen.

Mixture of Sodium nitrite and Ammonium Sulphate
 any ammonium salt)

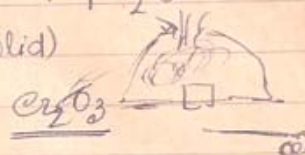




⑥ Ammonium Dichromate - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ [orange colour] solid

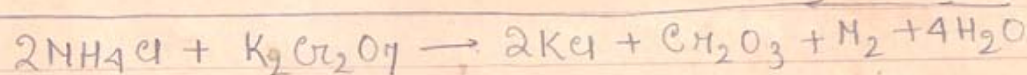
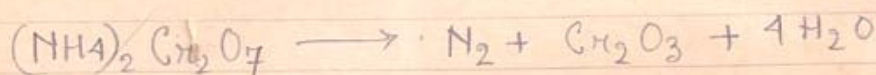
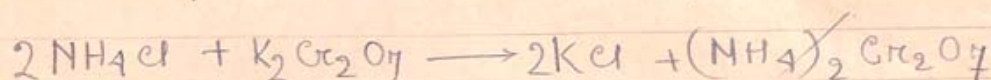


Volcano Experiment.

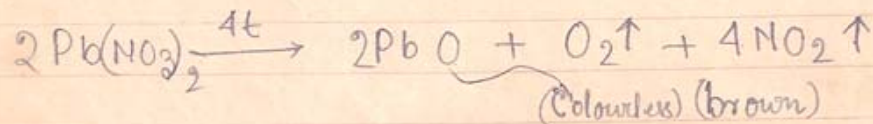


⑦

A Mixture of NH_4Cl and $\text{K}_2\text{Cr}_2\text{O}_7$



⑧ Lead nitrate: $\text{Pb}(\text{NO}_3)_2$



N.B A crackling sound is observed and brown gas evolved.

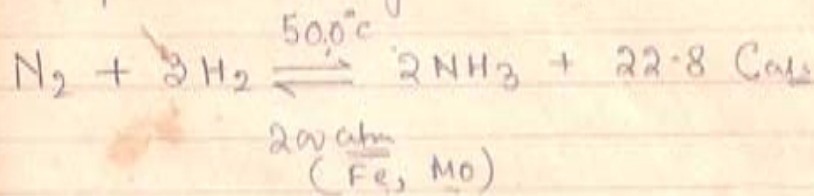
The parallel cases for $\text{Ba}(\text{NO}_3)_2$, $\text{Cu}(\text{NO}_3)_2$, $\text{Zn}(\text{NO}_3)_2$ etc.

Nitrogen



2) Haber Synthesis for the preparation of NH_3 industrially.

Theory: When a mixture of 3 vols of H_2 and 1 volume of Nitrogen is compressed under 200 atmospheric pressure and heated at nearly 500°C in the presence of finely divided Iron as a catalyst and a little molybdenum (Mo) as a promoter we get NH_3 .



Pressure is essential because when 4 vols of gas mixture is changed to 2 vol. Since the reaction is exothermic and reversible much heat is not needed. The optimum temp is 500°C .

According to Le Chatelier's principle the above conditions are favourable to get good yield.



Proce: Nitrogen & Hydrogen combines under the above conditions to give NH_3 . The resulting gases consist of NH_3 and uncombined N_2 and H_2 are cooled under pressure. When NH_3 is liquified and is taken out. The unreacted N_2 and H_2 are



mixed with fresh gases mixture and allowed to react in the catalytic chamber.

NITRIC ACID

Q7

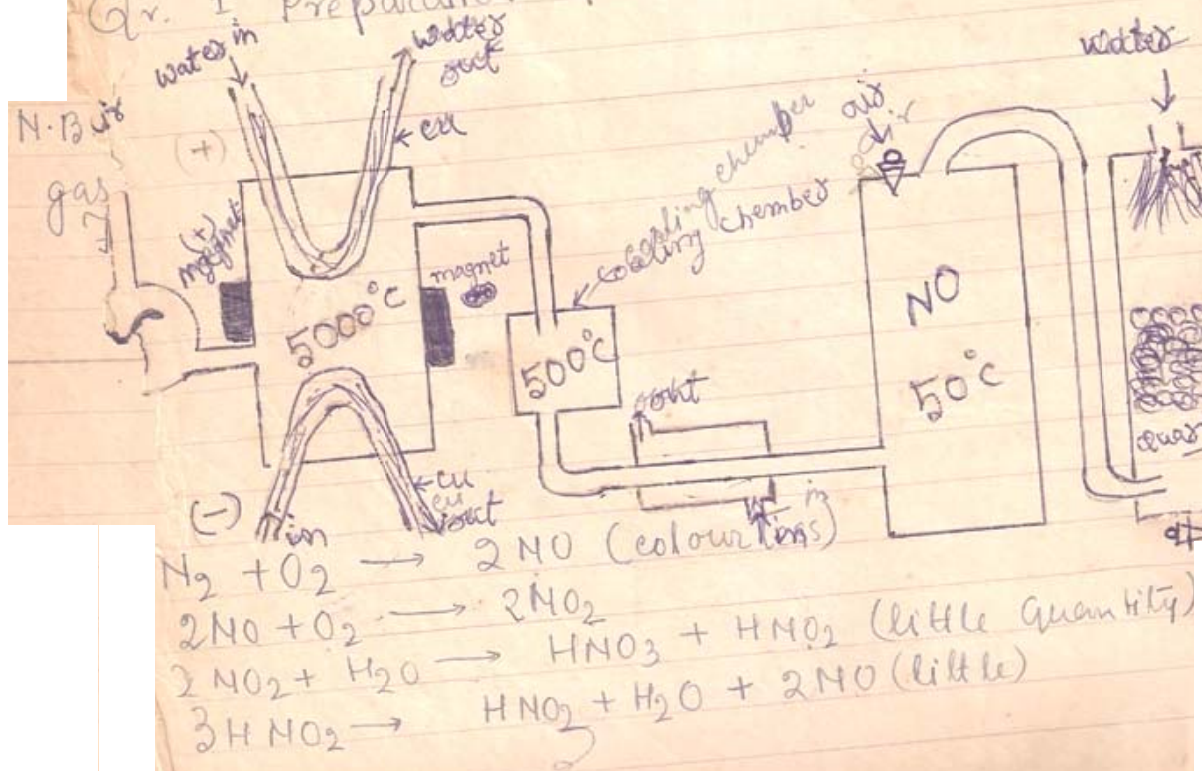
How is HNO_3 prepared in industry from atmosphere by Birkeland and Ede process or Aue process. Discuss the reaction of HNO_3 with

- (1) Non metals at least four.
- (2) Metals at least four.

Q. 2.

Prove that HNO_3 acid contains N_2 , O_2 and H_2 . Write notes on ring test and Aqua regia. Describe the conversion of NH_3 into nitric acid in industrial scale.

Q. 1 Preparation of HNO_3 :



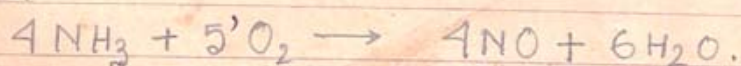
Nitrogen



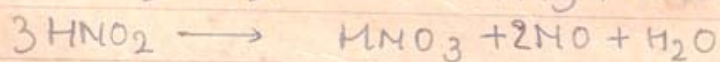
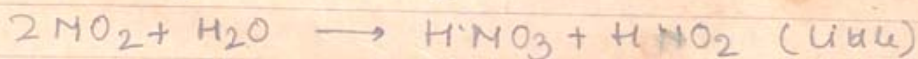
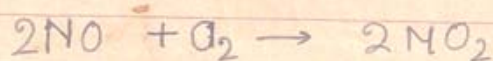
Procedure: Air, free from moisture and CO_2 is passed by a blower through an electric spark / arc. between two electrodes of copper tube through which cold water continuously flows. To spread the arc a horseshoe magnet is placed at right angle to the electrodes. N_2 & O_2 of air combine to form NO . It is allowed to cool in cooling chamber and again cooled to nearly room temperature by boiler. It is mixed with fresh air to convert into NO_2 , this NO_2 is then allowed to react with water in the absorption tower. By this method dilute HNO_3 is formed.

Industrial preparation of HNO_3 from NH_3 .
(Ostwald Process).

Theory: Ammonia is oxidised to NO by the Oxygen of air in the presence of heated Pt wire gauze, At about 700°C .

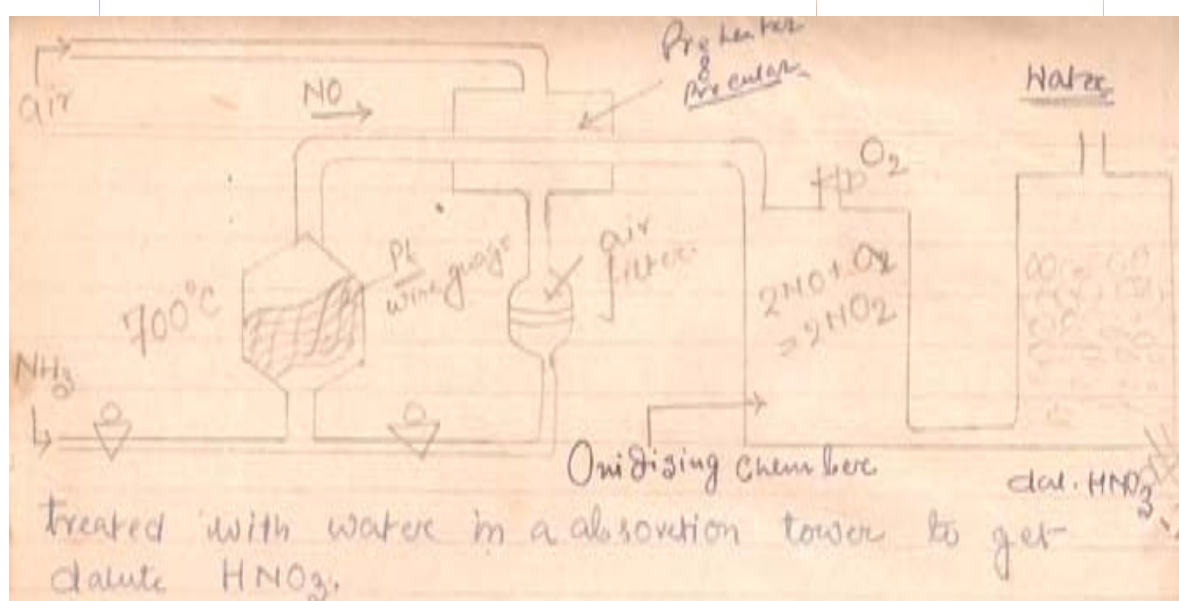


This NO with further Oxygen gives NO_2 which with H_2O gives HNO_3 .



A mixture of 1 vol. NH_3 and 7.5 vol. of air free from dust is very rapidly passed through a heated tube containing Pt wire gauze (Catalyst) at 700°C . NH_3 reacts with O_2 of air to form NO which with more O_2 in oxidising chamber gives NO_2 . This gas is now

Nitrogen



Concentration / Purification of HNO₃:

HNO₃ is purified by distilling fractionally in a glass retort.

(1) first fraction $\frac{1}{3}$ containing Cl, NO₂, water etc. is rejected.

(2) The second fraction $\frac{1}{3}$ is taken which is consider pure HNO₃ nearly 99.9%.

(3) The 3rd fraction $\frac{1}{3}$ left in the retort is rejected.

Sometimes a brown colour is observe, to decolourise it CO₂ gas is blown into it.

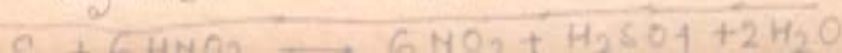
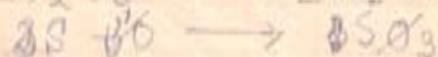
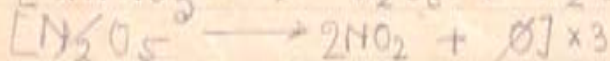
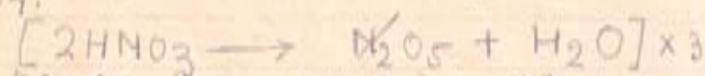
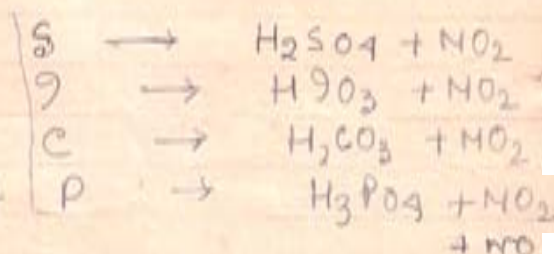
Properties of HNO₃:

With non metals:

81

(1) (a) With Sulphur:

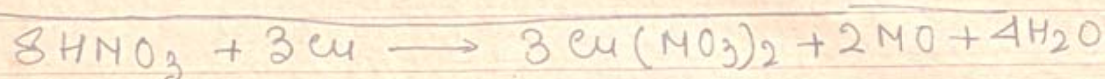
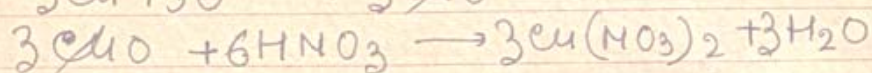
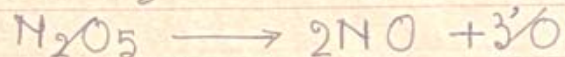
In boiling HNO₃ & S we get H₂SO₄.



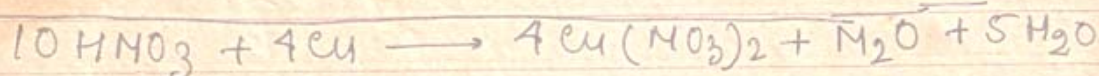
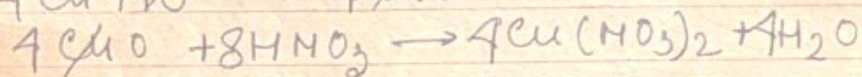
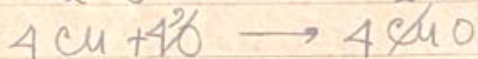
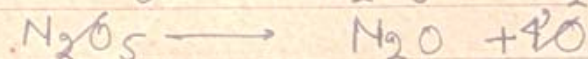
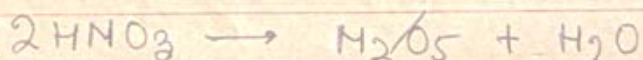
Nitrogen



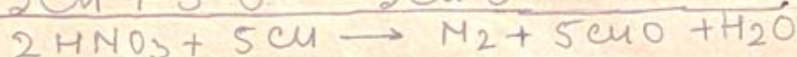
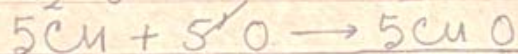
(b) With 50% HNO_3 :



(c) With 25% HNO_3 :

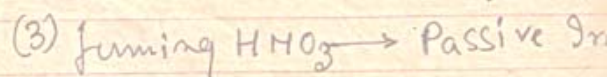
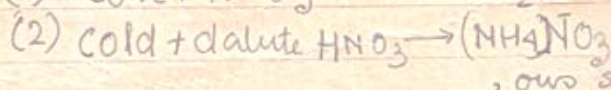
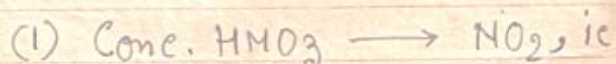


(d) With ^{vapour} ~~vapour~~ HNO_3 :



N.B: This is the prove that HNO_3 contain N_2

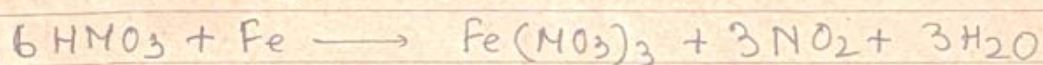
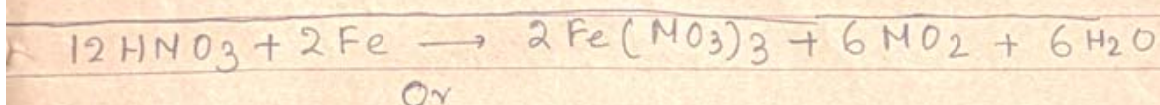
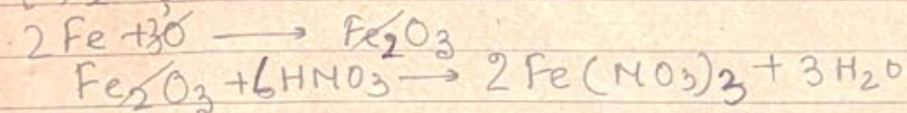
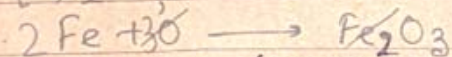
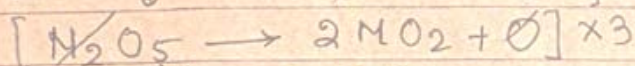
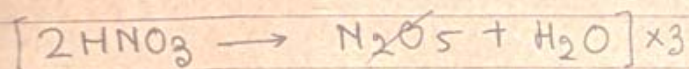
3. With Fe:



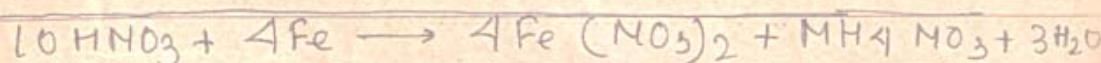
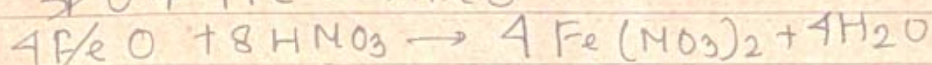
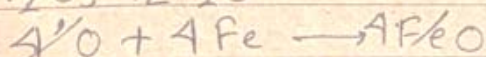
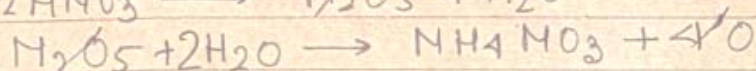


Nitrogen

With Conc. HNO_3 :



(b) With dil. HNO_3 :

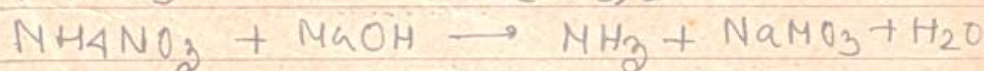
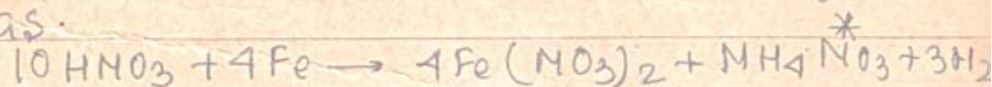


N:B Convert NH_3 to HNO_3 and vice versa

(a) $\text{NH}_3 \rightarrow \text{HNO}_3$ - Oswald Process discussed.

(b) $\text{HNO}_3 \rightarrow \text{NH}_3$:

Fe is treated with dilute HNO_3 to give NH_4NO_3 .
This product is boiled with NaOH to get NH_3 gas.





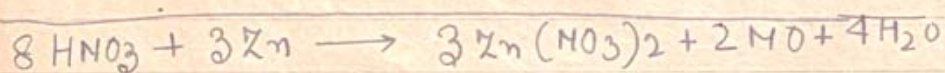
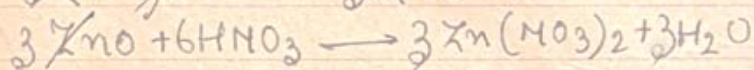
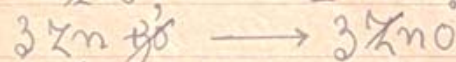
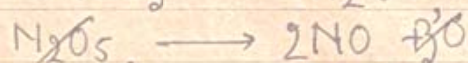
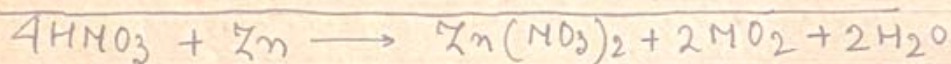
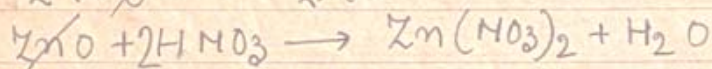
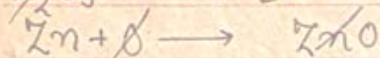
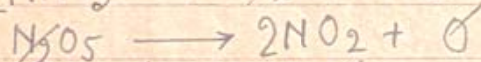
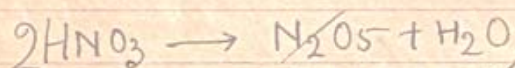
(c) fuming HNO_3 : (highly conc. HNO_3) When treated with Iron gives a coating or film of hypothetical oxide due to which Iron becomes passive. This passive iron when dipped in dilute HCl does not give H_2 but if it is rubbed with sand paper and put in dilute HCl it gives H_2 gas. This phenomenon of Iron is known as passivity of Iron.

4. With Zn :

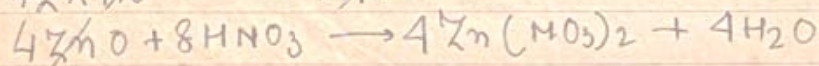
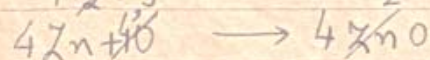
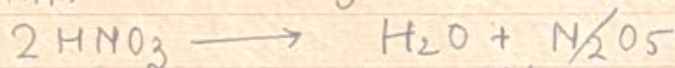
(a) With Cone HNO_3 - It gives NO_2 .

75%.	—	NO_2
50%.	—	NO
25%.	—	N_2O

Very very dilute - (NH_4)



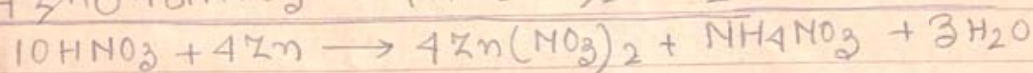
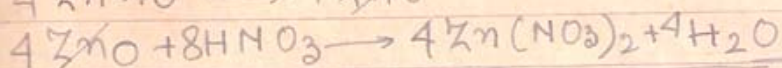
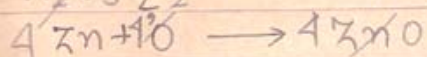
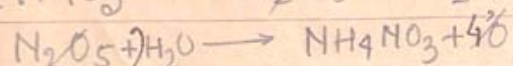
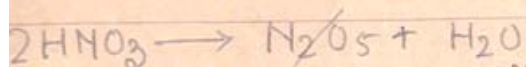
c) With 25% HNO_3 :





Nitrogen

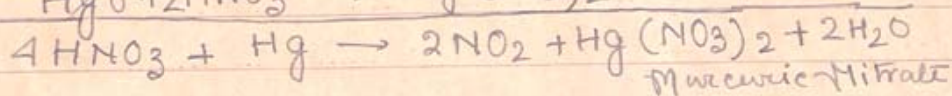
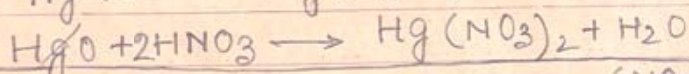
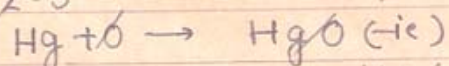
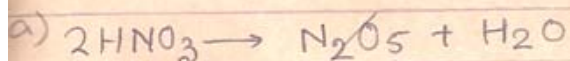
1) With very very dilute HNO_3 :



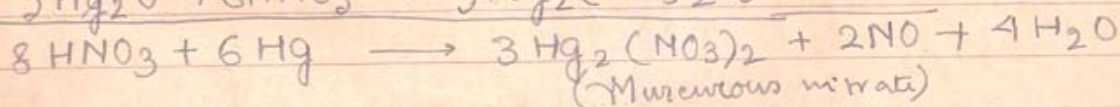
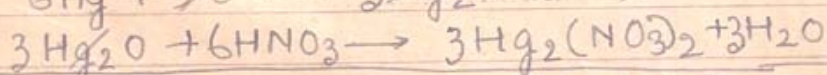
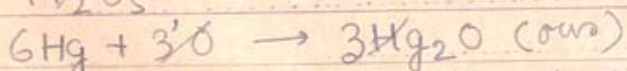
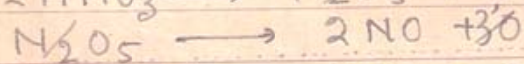
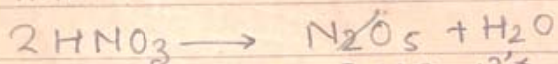
5. Reaction with Hg.

With Conc. HNO_3 :

Conc. HNO_3 — NO_2 — ie HgO
 dil. HNO_3 — NO — ous

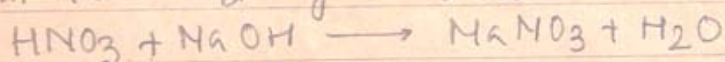


(b) With dilute HNO_3 :

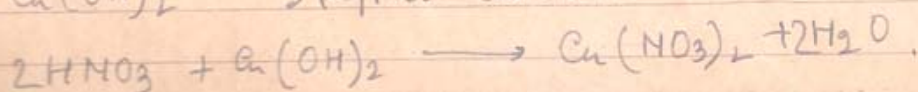


2. Reaction of HNO_3 with Alkali, showing that it is an acid.

1. With NaOH — It gives Sodium nitrate



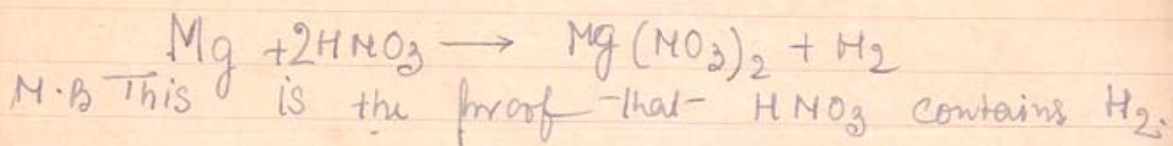
with $\text{Ca}(\text{OH})_2$ — It gives Calcium nitrate



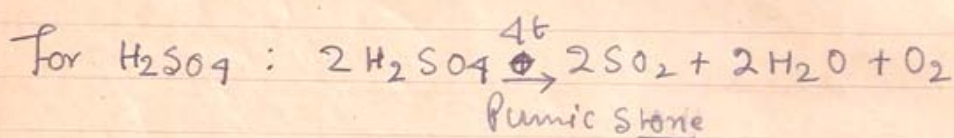
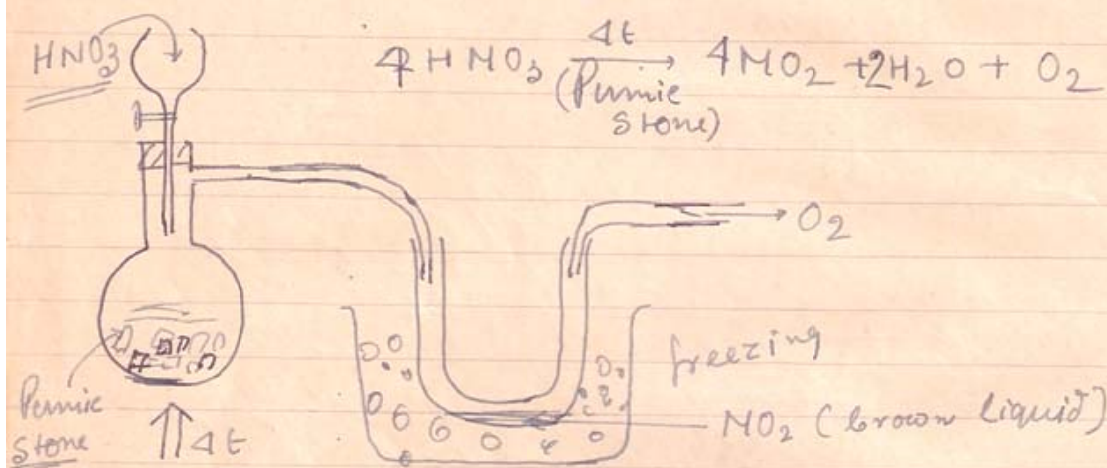


Nitrogen

3. With Mg/Mn-It gives H_2 gas



to prove HNO_3 contains O_2 :

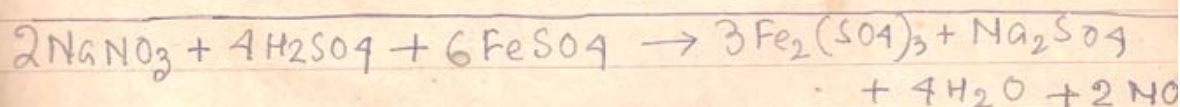
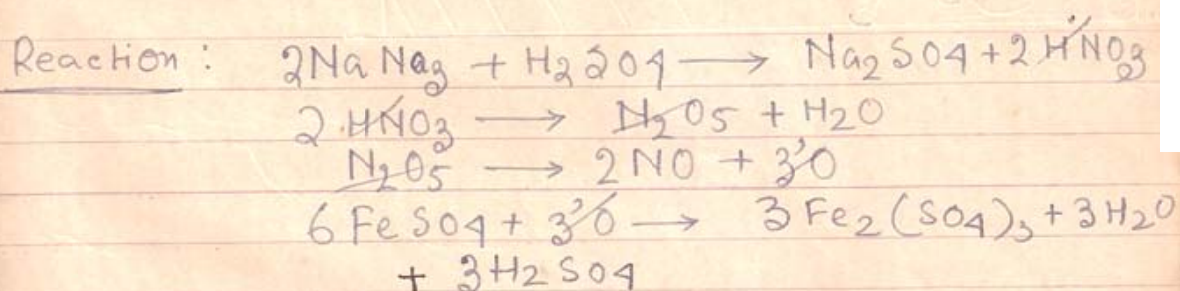
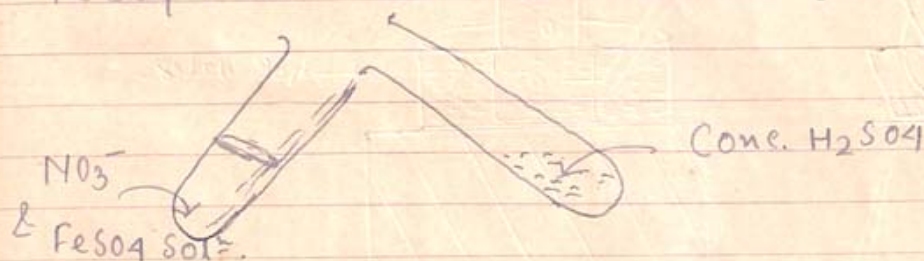


Con. HNO_3 is added to heated pumic Stone drop by drop to get a gas mixture of NO_2 (brown) and O_2 (colourless). The mixture is passed through U tube kept immersed in freezing mixture where brown gas is condensed to a liquid and the rest gas is nothing but Oxygen because it has similar properties with that of oxygen.



Ring test: Ring test is the confirmatory test for NO_3^- (nitrate) radical during Salt test.

Procedure: In a test tube nitrate salt is dissolved in water then added about equal volume of freshly prepared FeSO_4 solution and then added almost equal volume of Conc. H_2SO_4 from the side of the test tube so that H_2SO_4 will occupy the lower part. At the mixing point we get beautiful brown ring of $\text{FeSO}_4 \cdot x\text{NO}$.

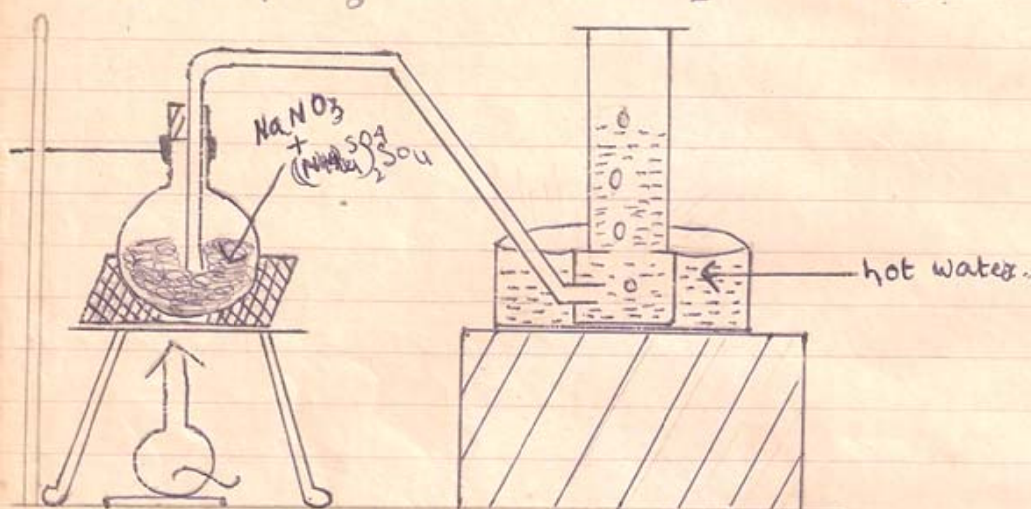
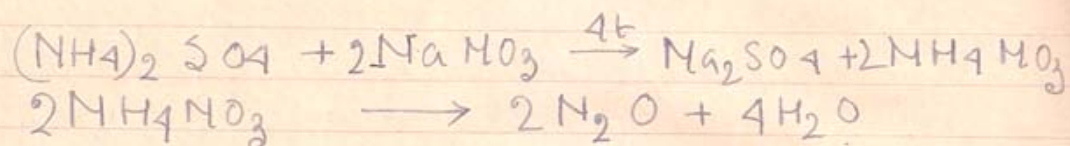


This NO gas is absorbed by FeSO_4 solⁿ causing brown ring having the formula $\text{FeSO}_4 \cdot x\text{NO}$ at the admixture point (mixing point) where x is integer.



Laughing Gas [N_2O]:

Preⁿ: It is prepared by heating a mixture of ammonium sulphate and sodium nitrate.



Procedure: In a round bottom flask about 30gms of $NaNO_3$ and 25gms of $(NH_4)_2SO_4$ is taken. By heating the mixture of two we get the vapour of nitrous oxide which is collected by the downward displacement of hot water, as shown in the figure above. As N_2O is slightly soluble in cold water so hot water is taken in place of cold water.

Purification: This gas may be purified by passing through washbottle containing respectively KOH (to remove NO_2 , $FeSO_4$ ~~to remove~~ remove NO , Conc. H_2SO_4 to remove water and then collected by displacement of mercury.

Precaution: Only NH_4NO_3 may be taken to get N_2O but to avoid explosion the above mixture is taken.

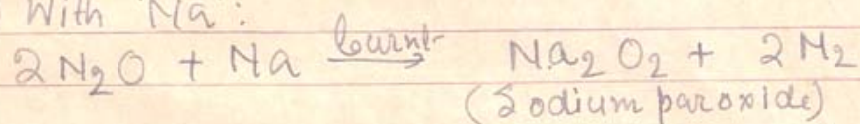


Nitrogen

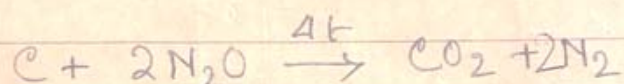
Physical Properties: It is a colourless gas slightly soluble in cold water but insoluble in hot water. It has a faint smell and if taken through nose (inhaled) it causes excitement accompanied by laughing hence the name is laughing gas. However if inhaled for a long time it causes insensibility to pain and may cause death.

Chemical properties:

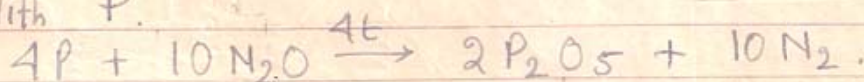
(1) With Na:



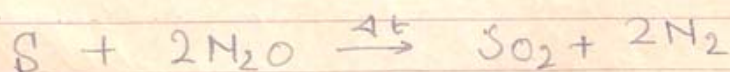
(2) With C



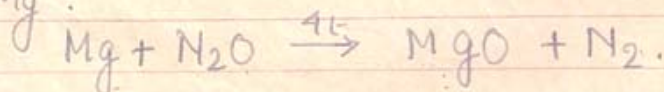
(3) With P.



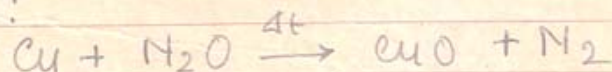
(4) With S.



(5) With Mg:



(6) With Cu:

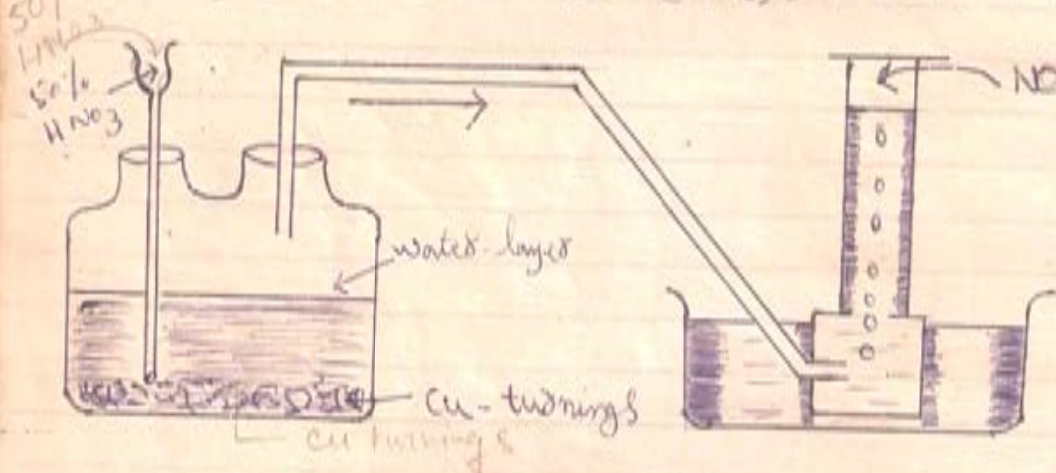
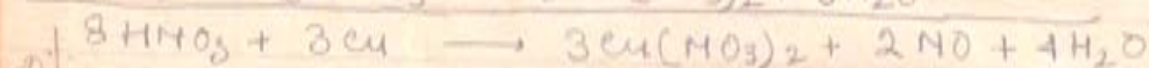
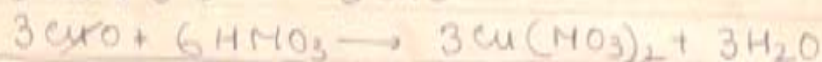
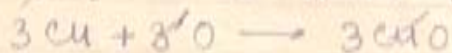
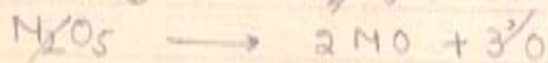
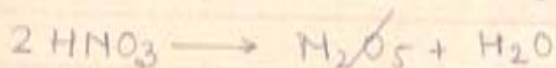




Nitrogen

Q. How is nitric oxide prepared in the lab. in fairly pure form. Discuss its chemical properties and mention the uses of it.

Theory: When 50% HNO_3 is treated with copper at ordinary temperature we get NO gas.



Procedure: In a two-necked bottle about 50 gms of Cu turnings is taken, fitted with thistle funnel up to the bottom of the bottle, fitted with delivery tube as shown in the figure. 50

50% HNO_3 / two test tube of conc HNO_3 and two test tubes of distilled water are added into it. At ordinary temperature reaction sets in (starts) to evolve NO gas. At the beginning the bottle is filled with brown gas due to the formation of NO_2 with O_2 present inside the bottle therefore the brown gas is



Nitrogen

allowed to escape and then the colourless NO gas is collected by the downward displacement of water.

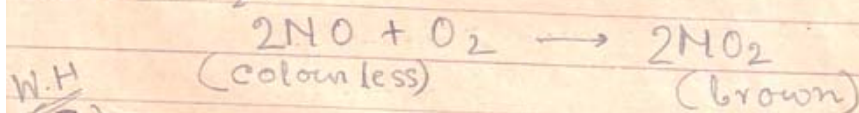
Precaution: As NO is a colourless gas, the brown NO₂ gas should be escaped and then NO is collected.

Purification - To the gas give FeSO₄ (saturated) solⁿ is added where by only NO gas is absorb giving a dark brown liquid having the formula FeSO₄·xNO. This dark brown liquid on heating gives pure NO.

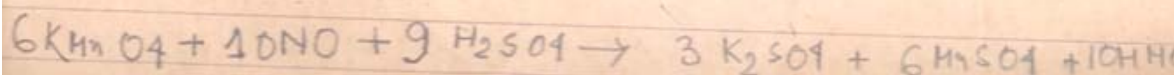
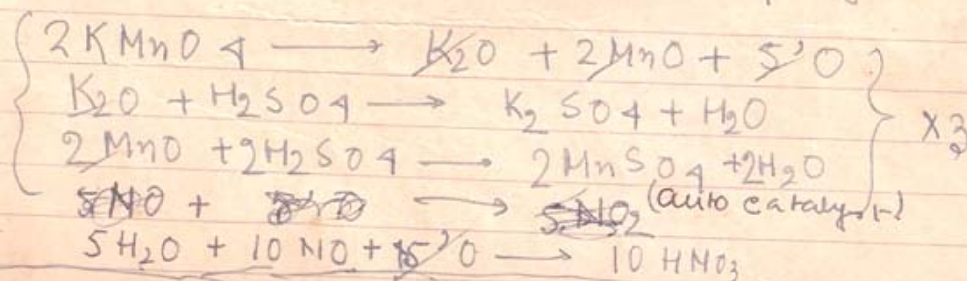
Properties: It is a colourless gas insoluble in water slightly heavier than air

Chemical properties:

(1) With O₂:



(2) With KMnO₄ solⁿ: When NO gas is bubbled through acidified KMnO₄ solⁿ the pink colour of KMnO₄ is discharged due to the formation of MnSO₄. Here it proves NO is a reducing agent.



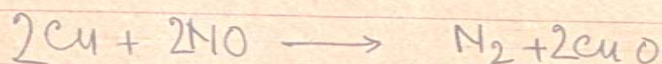


Nitrogen

(3) With H_2 - When NO and H_2 mixture is heated in the presence of Pt NH_3 is formed.



(4) With Heated Cu : When NO is passed over heated copper N_2 is set free (any oxide of H_2)



Uses: It is used as oxygen carrier in the preparation of H_2SO_4 in lead chamber process.